IBM DATA SCIENCE CAPSTONE PROJECT

PRESENTED BY :M. Shashank

PROJECT OVERVIEW

- **Objective**
- The objective of this project is to predict whether the first stage of a SpaceX Falcon 9 rocket will land successfully.
- **Why is this important?**
- Falcon 9 is a reusable launch vehicle.
- Successful landings reduce launch costs.
- Predicting landing success helps estimate the overall mission cost.
- Data science techniques can improve decision-making using historical launch data.

DATA COLLECTION

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
0	1	2006-03-24	None	None	None	None	None None	1	False	False	False	None	None	None	None
1	2	2007-03-21	None	None	None	None	None None	1	False	False	False	None	None	None	None
2	4	2008-09-28	None	None	None	None	None None	1	False	False	False	None	None	None	None
3	5	2009-07-13	None	None	None	None	None None	1	False	False	False	None	None	None	None
4	6	2010-06-04	None	None	None	None	None None	1	False	False	False	None	None	None	None

The launch data was collected from multiple sources:

SpaceX REST API

Wikipedia Web Scraping

Historical Falcon 9 launch records

Tools Used

Python

Requests

BeautifulSoup

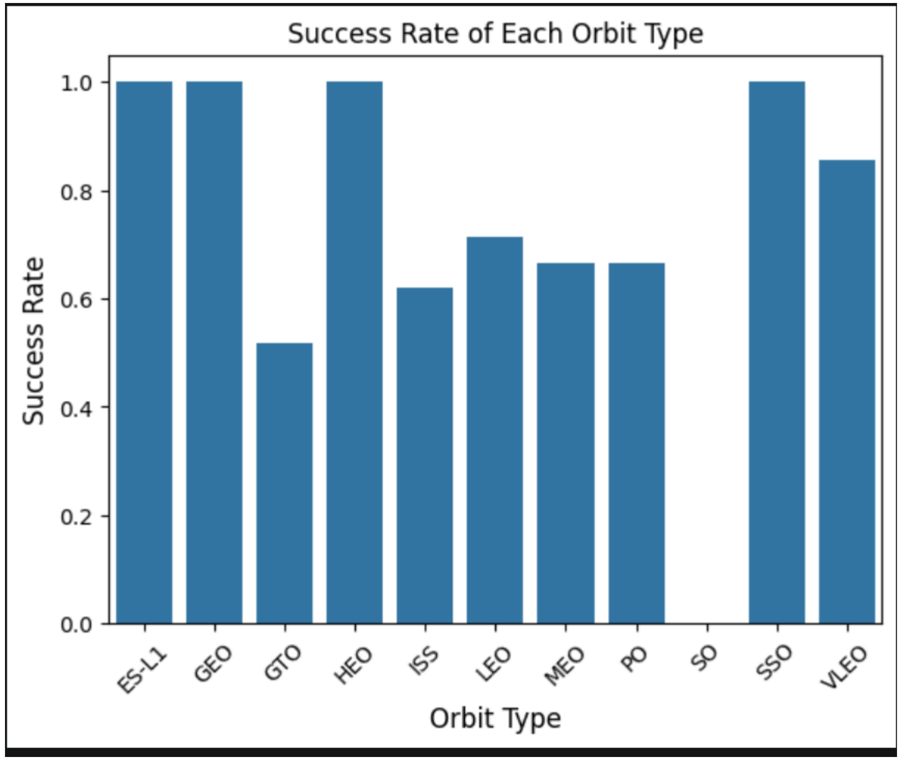
Pandas

DATA WRANGLING

```
FlightNumber       0.0
Date               0.0
BoosterVersion     0.0
PayloadMass        0.0
Orbit              0.0
LaunchSite         0.0
Outcome            0.0
Flights            0.0
GridFins           0.0
Reused             0.0
Legs               0.0
LandingPad         0.0
Block              0.0
ReusedCount        0.0
Serial             0.0
Longitude          0.0
Latitude           0.0
dtype: float64
```

- **Data Wrangling**
- The collected data was cleaned and prepared before analysis.
- **Steps Performed**
- Removed Falcon 1 launches
- Filtered only Falcon 9 launches
- Handled missing values
- Cleaned duplicate records
- Prepared the dataset for analysis
- **Result**
- A clean and structured dataset was created for further analysis and machine learning.

EXPLORATORY DATA ANALYSIS (EDA)



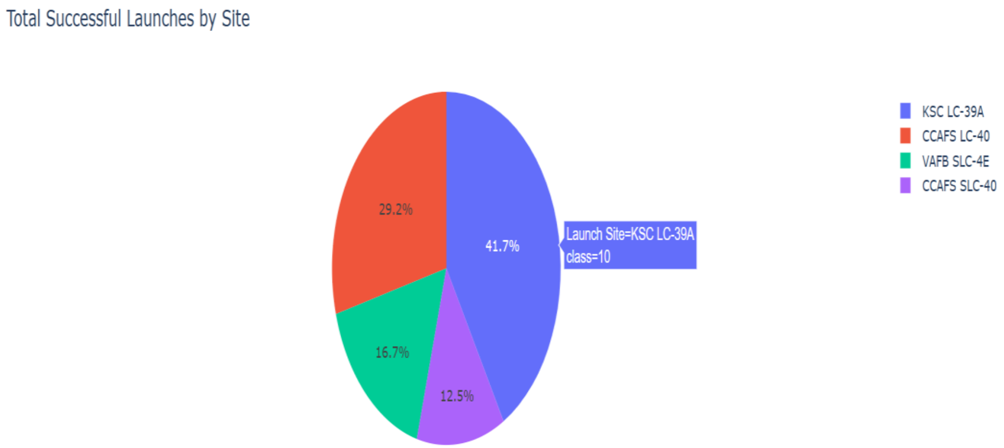
- **Exploratory Data Analysis**
- Several factors affecting landing success were analyzed.
- **Analysis Performed**
- Payload Mass vs Landing Success
- Flight Number vs Landing Success
- Orbit vs Landing Success
- Launch Site vs Landing Success
- **Observation**
- The analysis revealed patterns between launch characteristics and successful landings.

SQL ANALYSIS

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- **SQL Analysis**
- SQL was used to analyze the launch dataset.
- **Queries Performed**
- Total number of launches
- Launches by site
- Booster versions
- Payload statistics
- Successful missions
- **Result**
- SQL helped extract meaningful information from the dataset for further analysis.

INTERACTIVE DASHBOARD



- **Interactive Dashboard**
- An interactive dashboard was developed to visualize launch information.
- **Dashboard Features**
- Launch Site Selection
- Payload Range Slider
- Pie Chart
- Scatter Plot
- Interactive Filtering
- **Benefit**
- The dashboard enables users to explore launch data visually and interactively.

MACHINE LEARNING

```
Logistic Regression Test Accuracy: 0.8333333333333334
Support Vector Machine Test Accuracy: 0.8333333333333334
Decision Tree Test Accuracy: 0.8333333333333334
K-Nearest Neighbors Test Accuracy: 0.8333333333333334
The best performing model is Decision Tree.
```

- **Machine Learning Models**
- Several machine learning models were trained to predict landing success.
- **Models Used**
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbours (KNN)
- **Result**
- The models were evaluated using test accuracy, and the best-performing model was selected based on its performance.

CONCLUSION

- **Conclusion**
- **Project Summary**
- Successfully collected SpaceX launch data.
- Cleaned and prepared the dataset.
- Performed exploratory data analysis.
- Built an interactive dashboard.
- Developed machine learning models to predict Falcon 9 landing success.
- **Future Scope**
- Improve prediction accuracy using more advanced machine learning models.
- Include additional launch parameters.
- Develop a real-time prediction system.

THANK YOU 🙏

Questions?

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